FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 1HA-I 2308 — B(3084) / D(7084)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

Note: PHYSICS HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 3084. Fill the relevant bubble against each question according to curriculum:

Q1. Which one of the following is a dimensionless quantity?

(A) Strain
(B) Spring constant 'K'
(C) Young's constant 'Y'
(D) Stress

Q2. Real and apparent weights of a body seem to be equal when body moves with acceleration:

(A) $a > g$
(B) $a = g$
(C) $a = 0$
(D) $a < g$

Q3. Gravity performs zero work when body moves:

(A) Vertically
(B) At an angle of 30°
(C) At an angle of 60°
(D) Horizontally

Q4. Which one is TRUE for isothermal process?

(A) $\Delta W = 0$
(B) $\Delta Q = \Delta W$
(C) $\Delta U = \Delta W$
(D) $\Delta Q = 0$

Q5. If the temperatures of source and sink of a Carnot engine (having efficiency η_1) are each increased by 200K, then the efficiency η_2 will:

(A) Decrease
(B) Become 1
(C) Remain unaffected
(D) Increase

Q6. Due to which phenomenon different colours appear in soap film in sunlight?

(A) Diffraction of light
(B) Scattering of light
(C) Interference of light
(D) Dispersion of light

Q7. In Young's double slit experiment, the fringe spacing is:

(A) $d/(\lambda L)$
(B) $\lambda/(Ld)$
(C) $L/(\lambda d)$
(D) $\lambda L/d$

Q8. For constructive interference, Path difference, d = :

(A) $(m+1/2)\lambda$
(B) $(m-1/2)\lambda$
(C) $m\lambda^2$
(D) $m\lambda$

Q9. When length of a simple pendulum is doubled, the ratio of old to new time period will be:

(A) 1:1
(B) 1:2
(C) $1:\sqrt{2}$
(D) 2:1

Q10. If 30 waves pass through a medium in 1 second with speed of $30ms^{-1}$ then the wave length of waves is:

(A) 1m
(B) 2m
(C) 400m
(D) 20m

Q11. Stars moving away from the earth show:

(A) Blue shift
(B) Red shift
(C) Violet shift
(D) Black shift

Q12. If tension in a stretched string is made four times then velocity of waves:

(A) Becomes twice
(B) Becomes half
(C) Remains same
(D) Becomes four times

Q13. The dimension of spring constant K is:

(A) $[MT^{-2}]$
(B) $[ML^2]$
(C) $[MLT^{-1}]$
(D) $[MLT^2]$

Q14. The velocity of projectile at its maximum height is:

(A) v_iCosθ
(B) Maximum
(C) Zero
(D) v_iSinθ

Q15. If $\vec{A}=2i-j+3k$ then magnitude of $\vec{A}$ is:

(A) $\sqrt{14}$
(B) $\sqrt{18}$
(C) 14
(D) 4

Q16. $F.\Delta t$ = _____ :

(A) Time
(B) Pressure
(C) Impulse
(D) Force

Q17. In equation of continuity, the 'volume flow rate' is equal to:

(A) Ad
(B) Ap
(C) Av
(D) $A\Delta t$

Note: —1HA-I 2308—. Page 1 of 1. Formulae strip: $T=2\pi\sqrt{(l/g)}$; $v=f\lambda$; $A=\sqrt{(A_x^2+A_y^2+A_z^2)}$; $\eta=1-T_2/T_1$; $v=\sqrt{(T\times L/M)}$; $\varepsilon=\Delta L/L_o$; $\sigma=F/A$; $W=FS\ Cos\theta$; $T=2\pi\sqrt{(m/K)}$.

Note: PHYSICS HSSC–I. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any FOURTEEN parts from Section 'B' and attempt any TWO questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly.

Q. SECTION – B (Marks 42)

Q. 2 Answer any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) Differentiate between 'precision' and 'accuracy'.

(ii) How are cranes able to lift heavy load without toppling? Explain briefly.

(iii) Prove that the equation E = hf is dimensionally correct. (Where E = energy and f = frequency)

(iv) Co-relate Newton's third law of motion and conservation of momentum with the help of an example.

(v) What is impulsive force? Explain the effect of lengthening of time on impulsive force. Give one of its common life applications.

(vi) What will be the effect of gravity on vertical and horizontal components of projectile velocity? Give its mathematical expression(s).

(vii) Describe that work done is equal to area under the force-displacement graph.

(viii) What is escape velocity (v_{esc})? Derive the expression $v_{esc} = \sqrt{(2gR_e)}$.

(ix) Why is a rifle barrel "rifled"? Give reason in context of conservation of angular momentum.

(x) Briefly explain the following terms: (a) Moment of inertia of a body (b) Angular momentum.

(xi) Which object will attain terminal velocity first, a lighter object or a heavier object? Justify your answer.

(xii) A 550kg uniform I-beam supports a load of 320kg as shown. Determine the reactions at the supports. (Note: a diagram shows the beam with supports R_A and R_B, a 11.2 m span and a 4.8 m overhang carrying the 320 kg load.)

(xiii) In orbiting satellites artificial gravity is created to counter balance weightlessness. Determine the frequency 'f' required to produce this artificial gravity. (Note: a diagram shows a rotating satellite of radius r with tangential velocity v.)

(xiv) What will be the frequency of a simple pendulum if its length is 2 meters?

(xv) How progressive waves differ from stationary waves? Give examples also.

(xvi) Discuss first mode of vibration in a stretched string. Derive expression for its frequency. (Note: a diagram of the fundamental mode or first harmonic N-A-N is given.)

(xvii) What is meant by the path difference with reference to the interference of two waves?

(xviii) State Huygen's Principle. Use it to construct wave front after a time interval 't'.

(xix) Is the energy degraded during all natural processes? Explain.

(xx) Find Moon's angular momentum using Earth-Moon distance = 3.8×10^{8}m and Mass of Moon = 7.35×10^{22}kg.

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. Show that the potential at a point is equal to work done in bringing a unit mass "m" from infinity to that point. b. Differentiate 'molar specific heat at constant pressure (C_p)' and 'molar specific heat at constant volume (C_v)'. Justify that C_p > C_v. Also Prove that C_p – C_v = R

Q. 4 a. Describe S.H.M. Prove that the Projection of a body moving in a circular path executes S.H.M along the diameter. Also derive an expression for time period and frequency of this body. b. A source of sound and observer are moving toward each other. What happens to the apparent pitch heard by the observer? What will happen if both are moving away from each other? Explain and prove.

Q. 5 a. State and derive Bernoulli's equation. Also give one of its applications. b. Calculate the resultant of two forces of 30N and 40N acting at a point making angles of 0° and 120° with x-axis respectively.

> Note: Important formulae: $f=1/T$; $F=\sqrt{(F_x^2+F_y^2)}$; $T=2\pi\sqrt{(l/g)}$; $\vec{L}=\vec{r}\times\vec{p}$; $\vec{p}=m\vec{v}$; $V_t=2\rho g r^2/(9\eta)$; For equilibrium $\Sigma\tau_o=0$, $\Sigma F_x=0$, $\Sigma F_y=0$; Plank's Constant $h=6.626\times10^{-34}$JS; $F_R=\sqrt{(F_{Rx}^2+F_{Ry}^2)}$; $\theta=\tan^{-1}(F_{Ry}/F_{Rx})$; $J=F\times\Delta t$; $\vec{\tau}=\vec{r}\times\vec{F}$; $\Delta p=F\times\Delta t$; $V_t=m_g/(6\pi\eta r)$. —1HA-I 2308—

VERSION D

> Note: PHYSICS HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7084. Fill the relevant bubble against each question according to curriculum:

Q1. In Young's double slit experiment, the fringe spacing is:

(A) d/(λL)
(B) λ/(Ld)
(C) L/(λd)
(D) λL/d

Q2. The SI unit of product of pressure and volume is:

(A) Watt
(B) Pascal
(C) Newton
(D) Joule

Q3. According to first law of thermodynamics, which one is correct?

(A) C_p = 1 + R/C_v
(B) R = C_v/C_p
(C) C_p = R + C_v
(D) C_p + C_v = R

Q4. One radian = _____.

(A) 157.3°
(B) 90°
(C) 57.3°
(D) 180°

Q5. If $\vec{A}=5i-j$ then this vector makes an angle of ____ with positive x-axis.

(A) 449°
(B) 249°
(C) 349°
(D) 149°

Q6. A car is moving on a motorway. Point out the correct velocity-time graph which shows zero acceleration of car. (Note: four velocity-time graphs are given as options.)

(A) Option A (v-t graph: velocity decreasing to zero)
(B) Option B (v-t graph: horizontal line)
(C) Option C (v-t graph: rising curve)
(D) Option D (v-t graph: rising straight line)

Q7. A stone is thrown to perform projectile motion. Which one of the following is TRUE for its vertical acceleration?

(A) Constant
(B) Maximum at highest point only
(C) Maximum at the point of projection only
(D) Zero

Q8. Gravity performs zero work when body moves:

(A) Horizontally
(B) At angle of 30°
(C) At angle of 60°
(D) Vertically

Q9. Power, P = ____ :

(A) $\vec{F}.\vec{v}$
(B) $\vec{F}/\vec{A}$
(C) $\vec{F}/\vec{v}$
(D) $\vec{F}.\vec{d}$

Q10. The ratio of angular momentum and angular velocity is equal to:

(A) Moment of inertia
(B) Angular acceleration
(C) Torque
(D) Mass

Q11. Real and apparent weights of a body seem to be equal when:

(A) $a = 0$
(B) $a < g$
(C) $a > g$
(D) $a = g$

Q12. The fluid speed is measured by:

(A) Barometer
(B) Hydrometer
(C) Monometer
(D) Venturi meter

Q13. If the period of oscillation of mass M suspended from a spring is 2S, then the period of mass 16M will be:

(A) 2S
(B) 4S
(C) 8S
(D) 1S

Q14. The total energy of a particle executing S.H.M is proportional to:

(A) Frequency of oscillation
(B) Velocity of particle
(C) Square of amplitude of motion
(D) Displacement from mean position

Q15. The fundamental frequency of an open organ pipe is 'f'. What will be fundamental frequency if its one end is closed?

(A) 1f
(B) 2f
(C) 3f
(D) 0.5f

Q16. The speed of sound in air does not depend upon:

(A) Density
(B) Humidity
(C) Temperature
(D) Pressure

Q17. Due to which phenomenon different colours appear in soap film in sunlight?

(A) Diffraction of light
(B) Scattering of light
(C) Interference of light
(D) Dispersion of light

> Note: —1HA-I 2308 HA—. Page 1 of 1. Formulae strip: $\theta=\tan^{-1}(F_y/F_x)$; $T=2\pi\sqrt{(m/K)}$; $2\pi\text{rad}=360°$; $L=I\omega$; $(K.E)_{max}=1/2\ kx_o^2$; $f_n=(2n-1)v/(4L)$; $f_{closed}=V/(4L)$; $f_{open}=V/(2L)$; $v=\sqrt{(E/\rho)}$; $f_n=nv/(2l)$; $v=\sqrt{(\gamma RT/m)}$.

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Q. SECTION – B (Marks 42)

Q. 2 Answer any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) Describe the terms 'error' and 'uncertainty' in context of measurements.

(ii) The length and width of a mobile phone is (8.30±0.01)cm and (4.60±0.01)cm respectively. Calculate the area of mobile phone and uncertainty in area.

(iii) Prove that the magnitude of vector product of two vectors is equal to area of Parallelogram.

(iv) In simple harmonic motion, at mean position velocity is maximum while acceleration is zero. Why?

(v) Differentiate between 'elastic' and 'inelastic' collisions with examples.

(vi) At what angle a stone is thrown such that its horizontal distance and vertical height become equal?

(vii) Calculate escape velocity for Mars if radius and acceleration due to gravity for Mars are 3.4×10^{6}m and 3.7m/s^{2} respectively.

(viii) If a satellite is in the state of free fall, then why does it not fall on the earth?

(ix) Yellow light of wavelength 5893×10^{-10}m is directed upon two narrow slits 0.20cm apart in Young's experiment. Find the position of the first bright and dark fringes on a screen 200cm away.

(x) Which wheel of the tractor shown in figure shall rotate faster when tractor moves with uniform velocity? Explain briefly. (Note: a diagram of a tractor with Front Wheel and Rear Wheel marked is given.)

(xi) Does the magnitude of viscous force in fluid flow depend on the shape and velocity of the object? Justify your answer.

(xii) When water falls from a tap its cross-sectional area decreases as it comes down. Why? (Note: a diagram of a tap with cross sections A_1V_1 and A_2V_2 marked is given.)

(xiii) Calculate acceleration of simple pendulum executing S.H.M with time period of 0.60S at displacement 0.05m.

(xiv) What will be the frequency of a simple pendulum if its length is 2 meters?

(xv) Discuss stationary waves in a stretched string when string is plucked at quarter length of string.

(xvi) What is meant by the path difference with reference to the inference of two waves?

(xvii) Prove $2d\sin\theta=m\lambda$, Bragg's law, for diffraction of X-rays through crystals.

(xviii) Is the energy degraded during all natural processes? Explain briefly.

(xix) Why do bowlers shine one side of cricket ball? Explain with reference to Bernoulli effect.

(xx) Describe the terms 'beat' and 'beat frequency'. How are beats useful for tuning the musical instruments?

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. What is meant by scalar product of two vectors? Express the scalar product of two vectors in terms of their rectangular components. Also prove that $\vec{A}.\vec{B} = \vec{B}.\vec{A}$. b. What is Newtons' formula for the speed of sound in the air? How did Laplace correct this formula? Explain in detail.

Q. 4 a. Show that the potential at a point is equal to work done in bringing a unit mass "m" from infinity to that point. b. Calculate and compare the velocities of Ring (hoop) and Disc (cylinder) at the bottom of inclined plane of height 'h'. (Note: a diagram of a ring and a disc rolling down an inclined plane of height h and angle θ is given.)

Q. 5 a. Differentiate 'molar specific heat at constant pressure (C_p)' and 'molar specific heat at constant volume (C_v)'. Justify that $C_p > C_v$. Also Prove that $C_p - C_v = R$. b. An object is at rest, a constant force acts on it and it starts moving with constant acceleration of 10m/s^2. How much distance will it travel in 3rd second of its Journey.

> Note: Important formulae: $v=r\omega$; $a=-\omega^2 x$; $v_{esc}=\sqrt{(2gR)}$; $R=v_o^2 Sin2\theta/g$; $H=v_o^2 Sin^2\theta/(2g)$; $\vec{A}.\vec{B}=ABCos\theta$; $T=2\pi/\omega$; $F_d=6\pi\eta r v_r$; $f=1/T$; $L=I\omega$; $y_{bright}=L\lambda/d$; $y_{dark}=L\lambda/(2d)$; $v_f=v_i+at$; $2aS=v_f^2-v_i^2$; $S=v_it+1/2\ at^2$; $\vec{L}=\vec{r}\times\vec{p}$; $\vec{p}=m\vec{v}$; $T=2\pi\sqrt{(l/g)}$. —1HA-I 2308 HA—